GHraisoni INTERNATIONAL SKILL TECH UNIVERSITY Pune

G H Raisoni International Skill Tech University, Pune.

Question Paper

Continuous Assessment Examination (CAE)-II

Programme: B Tech **Year: First** **Semester: II**
Max Marks: 20 **Duration: 60 mins.**
Course Name: Digital Electronics **Course Code: 24BTCL203)**

Instructions:

1. You are permitted to use calculators during the exam.
2. Clearly label each question and ensure your answers are organized.
3. Students must label their answers with the corresponding question number, such as Q.No. 1A, Q.No. 2C, etc.

Note: This question paper addresses below Course Outcomes (CO's):

CO1	To Demonstrate a clear understanding of binary number system, Boolean algebra, and the principles of digital logic
CO2	Ability to design, analyze, and optimize combi-national and sequential logic circuits using logic gates and Boolean algebra.
CO3	Understand the operation of memory elements, registers, and memory systems in digital electronics and its conversion, and apply this knowledge in practical scenarios

Q. No.	Sub Q. No.	Question	Marks
1		Solve any **TWO**	
	A	I. Explain Flip Flops in Digital Electronics II. What are the key applications of flip-flops in digital circuits?	05
	B	I. Explain the working of S-R flip-flops and its truth table. II. Explain the working of D flip-flop and its truth table.	05
	C	I. What is a shift resister and tis type? II. Explain working of 5-bit SISO with data flow with input 11111.	05
	D	Design 3-Bit Down Asynchronous Counter using J-K Flip Counter where Triggering is Positive.	05
2		Solve any **TWO**	
	A	I. Explain the working of a weighted resistor Digital-to-Analog Converter (DAC).. II. Find analog equivalent of 101010 (0 1 2 3 4 5) using weighted resistor DAC.	05
	B	Explain DAC Specifications.	05
	C	Differentiate between Analog to DAC and ADC. Write and explain at least 5 Points	05
	D	I. Discuss read and write operations in memory organization. II. Design a memory of size 16 × 4.	05

--- End of the Question Paper ---